

I. RESEARCH INTERESTS

Interested in atmospheric physics, weather dynamics, and climate science. Focus on using numerical models, observational data, and machine learning to study wind patterns, wave propagation, and monsoon processes.

II. EXPERIENCE

🔬 **India Meteorological Department (IMD)** **2025 – Present**
 Junior Research Fellow Pune, India
 • Researching atmospheric waves, mountain-forced winds, and weather patterns over complex terrains.
 • Using machine learning and Python to analyze weather and climate datasets.

III. EDUCATION

🎓 **Dr. B R Ambedkar National Institute of Technology** **Aug 2022 – May 2024**
 MSc in Physics Jalandhar, India

 🎓 **St. Aloysius College (Autonomous)** **June 2019 – July 2022**
 BSc in Physics, Chemistry, Mathematics Mangalore, India

IV. RESEARCH PROJECTS¹

🔬 **Numerical Modeling of Advective Boundary Layer Dynamics** **May 2026 – May 2026**
 ICTS Monsoon School Project Bengaluru, India
 • Built a numerical solver for the Mahrt (1972) boundary layer model using Python.
 • Initialized the model with ERA5 wind data along the 60°E longitude (Somali Jet axis).
 • Simulated wind profiles and studied boundary layer sensitivity to pressure forces and eddy viscosity.

🔬 **Autoencoder-based Precision Measurement Techniques** **July 2025 – Nov 2025**
 Supervisor: Dr. Raman Khurana Northwestern University (USA)
 • Built autoencoder neural networks to improve precision in particle physics measurements.
 • Used the models to calibrate measurements and reduce systematic uncertainties.

🔬 **Medium Modification of Pentaquark States** **July 2023 – May 2024**
 Master's Thesis Research NIT Jalandhar, India
 • Studied exotic pentaquark particles in dense stellar matter using quark field models.
 • Calculated mass modifications in nuclear and strange matter relevant to neutron star cores.
 • Used scalar and vector meson frameworks for dense matter simulations. [Link to thesis excerpts.](#)

🌐 **Galaxy Morphology Classification using Deep Learning** **May 2023 – July 2023**
 Independent Research Project
 • Built a convolutional neural network (CNN) to classify galaxy shapes using SDSS image data.
 • Created a data preprocessing and augmentation pipeline for astronomical images.
 • Evaluated classifier performance across multiple morphological classes. [Project details.](#)

V. TRAINING / SUMMER SCHOOLS

🎓 **International Centre for Theoretical Sciences (ICTS)**
 Moist Convective Dynamics of Monsoons - II Bengaluru, India; May 2026

¹Scan the QR code or visit the website for project codes and details: <https://mandeepairborne.github.io/>



Brown University Winter AI School

AI trends and applications in scientific computing

Rhode Island, USA; Winter 2025

Curves and Surfaces

Differential geometry and applications in theoretical physics

ICTS, Bengaluru, India; Summer 2022

Physics Training and Talent Search (PTTS)

Advanced undergraduate physics enrichment program

IISER TVM, India; Summer 2021

VI. SELECTED COURSES

- **Master's Courses:** Quantum Mechanics, Relativistic Quantum Mechanics, Particle Physics, Nuclear Physics, Electrodynamics, Statistical Mechanics, Mathematical Methods.
- **BSc Courses:** Classical Mechanics, Modern Physics, Thermodynamics, Distribution Theory, Linear Programming & Optimization.
- **Meteorology & Additional Courses:** Dynamics of Meteorology, Physical Meteorology, Machine Learning with Python, Deep Learning Specialization.

VII. AWARDS & ACHIEVEMENTS

- **CSIR-NET (Physical Sciences) – AIR 143:** Qualified for Assistant Professor appointment and PhD admission (Dec 2024).
- **IIT JAM – AIR 2283:** Qualified joint entrance exam for Master's programs (India).

VIII. SKILLS

- **Programming & Scripting Languages:** Python, C++, Bash, LaTeX
- **Tools & Libraries:** ROOT, SciPy, NumPy, Pandas, Matplotlib, TensorFlow, Keras, Mathematica
- **Operating Systems & Software:** Linux (Ubuntu & Arch), Git, Jupyter Notebooks, Overleaf

IX. LEADERSHIP & STUDENT GOVERNANCE

Department of Physics, NIT Jalandhar

Postgraduate Student Council Representative

India; 2023 – 2024

- Represented postgraduate physics students in the Institute Senate on academic curriculum and policy reform.

National Cadet Corps (NCC) -- Air Wing

C-Certificate Cadet

St. Aloysius College, Mangalore; 2019 – 2022

- Participated in leadership camps and studied aviation physics and basic aerodynamics.

X. REFERENCES

Dr. Soumi Chakravorty (PhD Supervisor)

Scientist-D

India Meteorological Department, Pune

Email: soumichakravorty.imd@gmail.com

Scholar Profile: [Google Scholar](#)

Dr. Suneel Dutt (Master's Thesis Supervisor)

Assistant Professor

Department of Physics, NIT Jalandhar, India

Email: dutts@nitj.ac.in

Scholar Profiles: [NIT Jalandhar](#) | [Google Scholar](#)

